**THE ABERDARE TECHNICAL COLLEGE**

**ICT TECHNICIAN LEVEL 5**

**CONTINUOUS PRACTICAL TEST 2 MOD 2**

**UNIT: COMPUTER NETWORK SET UP**

**UNIT CODE: ISCED UNIT CODE: 0612 351 03A**

**TIME: 1 HOUR 30 MINUTES**

**TOTAL MARKS: 50 MARKS**

**NAME: ________________________________________**

**REG NO: _____________________________________**

**DATE: ________________________________________**

## INSTRUCTIONS TO THE CANDIDATE

- ✓ *Attempt all tasks.*
- ✓ *Observe all networking laboratory safety measures throughout the practical.*
- ✓ *Use the provided networking devices, cables, and tools correctly.*
- ✓ *Demonstrate each completed task to the assessor before proceeding.*
- ✓ *Record all observations and results where required.*
- ✓ *Marks will be awarded for correct procedures, accuracy, and successful completion of tasks.*

# MATERIALS

- Router
- Switch (8-port or 24-port)
- 3 Desktop computers (PC1, PC2, PC3)
- UTP Cat5e/Cat6 network cables
- RJ45 connectors
- Patch cables (straight-through)
- Cable tester
- Multimeter
- Tape measure
- Crimping tool
- Wire stripper and cutter

## TASK 1: NETWORK CABLE TESTING

You have been provided with a network cable and cable testing tools.

### Practical Instructions

### (a) Continuity Testing

1. Connect both ends of the network cable to a cable tester.
2. Perform a continuity test.
3. Record whether the cable passes or fails the test.
4. Present the results to the assessor.

**Result:** ___________________________________________

### (b) Wire Mapping

1. Use the cable tester to verify the wire sequence.

2. Check that pins 1 to 8 are correctly mapped.
3. Record the test results.
4. Present the results to the assessor.

**Result: ____________________________________________**

**(c) Fault Detection**

1. Inspect the cable for faults.
2. Use a cable tester or multimeter to identify any fault.
3. Determine the type of fault detected.
4. Record your findings.

**Fault Identified: ___________________________________**

**(d) Cable Length Testing**

**Using the same cable provided by the assessor, measure and record the measurements**

1. Use tape measure to determine the cable length.
2. Record the measured length.
3. Present the results to the assessor.

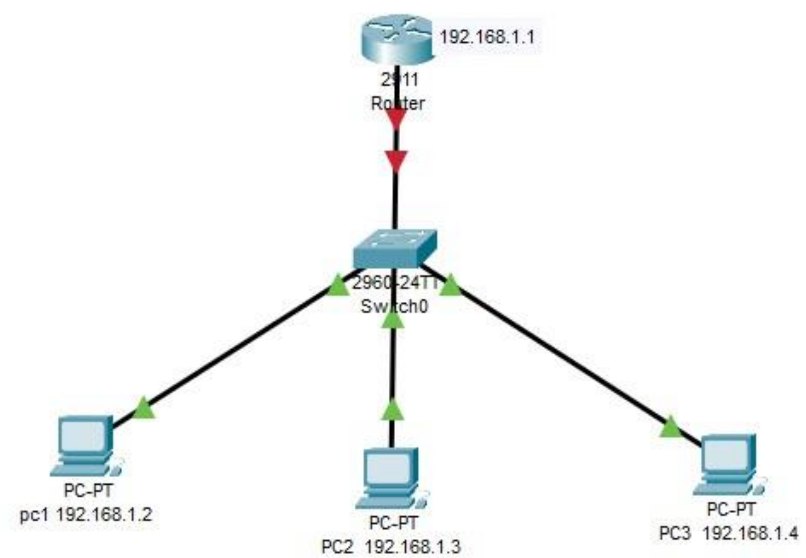
**Cable Length: _______________________________________**

## TASK 2: NETWORK DEVICE SETUP AND IP CONFIGURATION

You have been provided with the following equipment:

- 1 Router
- 1 Switch
- 3 Workstations (PC1, PC2 and PC3)
- UTP Network Cables

### Network Topology




### IP Addressing Scheme

| Device | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| Router | 192.168.1.1 | 255.255.255.0 | N/A |

| | | | |
|---|---|---|---|
| PC1 | 192.168.1.2 | 255.255.255.0 | 192.168.1.1 |
| PC2 | 192.168.1.3 | 255.255.255.0 | 192.168.1.1 |
| PC3 | 192.168.1.4 | 255.255.255.0 | 192.168.1.1 |

## Practical Instructions

1. Connect the Router to the Switch using a network cable.
2. Connect PC1, PC2 and PC3 to the Switch using network cables.
3. Connect the Access Point to the Switch.
4. Power on all devices.
5. Configure the IP addresses according to the addressing scheme provided above.
6. Verify that all cable connections are secure.
7. Open Command Prompt on each workstation and verify the IP configuration using the appropriate command.
8. Test connectivity between:
   - PC1 and PC2
   - PC1 and PC3
   - PC2 and PC3
   - PC1 and Router
9. Record the results of the connectivity tests.
10. Demonstrate the completed setup to the assessor.

## Connectivity Test Results

| Source Device | Destination Device | Result (Successful/Failed) |
|---|---|---|
| PC1 | PC2 | |
| PC1 | PC3 | |
| PC2 | PC3 | |
| PC1 | Router | |

**TASK 3: IDENTIFICATION OF NETWORK TOOLS**

Identify the following tools and state one use of each.

| Tool | Use |
|---|---|
| Cable Tester | |
| Multimeter | |
| Crimping Tool | |
| Wire Stripper/Cutter | |

**The trainer is introducing oral questions for one hour.**